higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

12 April 2024

This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1 D
1.2 B
1.3 A
1.4 A
1.5 D

(5 × 2) **[10]**

QUESTION 2

2.1 True
2.2 False
2.3 True
2.4 True
2.5 True
2.6 False
2.7 False
2.8 True

(8 × 1) **[8]**

QUESTION 3

3.1 E
3.2 F
3.3 G
3.4 D
3.5 A
3.6 C
3.7 B

(7 × 1) **[7]**

TOTAL SECTION A: 25

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SECTION B

QUESTION 4

4.1

MIN			1	0	1	0,	1	0	1	
MIN			1	1	0	0,	1	0	0	
					BINARY					
		1*	0	1	0	1	0,	1	0	1
		1*	0	1	1	0	0,	1	0	0
					1'S					
		1*	1	0	1	0	1,	0	1	0	√
		1*	1	0	0	1	1,	0	1	1	√
	1	1*	0	1	0	0	0,	1	0	1	√
										1	
		1*	0	1	0	0	0,	1	1	0	√
neg			1	0	1	1	1,	0	0	1	√

(5)

4.2

					BINARY							
POS			1	1	0	1,	0	1	1	1	1	
MIN			1	1	1	0,	1	0	0	1		
					BINARY							
POS	0*	0	1	1	0	1,	0	1	1	1	1	
MIN	1*	0	1	1	1	0,	1	0	0	1	0	
					2'S							
POS	0*	0	1	1	0	1,	0	1	1	1	1	
MIN	1*	1	0	0	0	1,	0	1	1	1	0	√
0	1*	1	1	1	1	0,	1	1	1	0	1	√
		0	0	0	0	1,	0	0	0	1	0	√
PLUS 1											1	
NEG						1,	0	0	0	1	1	√ √

(5)

4.3

						1	0	1	0	0,	1	1	0	1		
MULT									1	0	1,	0	1	1		
						1	0	1	0	0,	1	1	0	1		√
					1	0	1	0	0,	1	1	0	1			√
			1	0	1	0	0,	1	1	0	1					√
PLUS	1	0	1	0	0,	1	1	0	1							√
EQUAL	1	1	0	1	1	1	1,	1	1	0	1	1	1	1		√

(5)

4.4

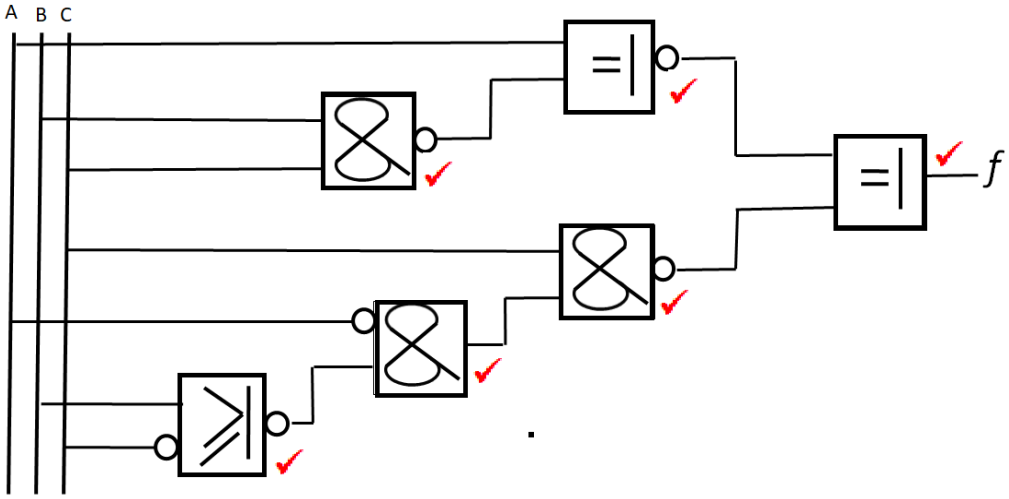
									1	0	1	0	0,	1	1	0	1 √
1	0	1,	0	1	1	1	1	0	1	1	1	1,	1	1	0	1	1	1	1
						1	0	1,	0	1	1								
								1	1	0	0	1	1						√
								1	0	1,	0	1	1						
										1	0	0	0	1	0	1			√
											1	0	1,	0	1	1			
										0	1	1	0	1	0	1			√
											1	0	1,	0	1	1			
													1	0	1	0	1	1	√
													1	0	1,	0	1	1	
																		0	

(5)
[20]

QUESTION 5

5.1

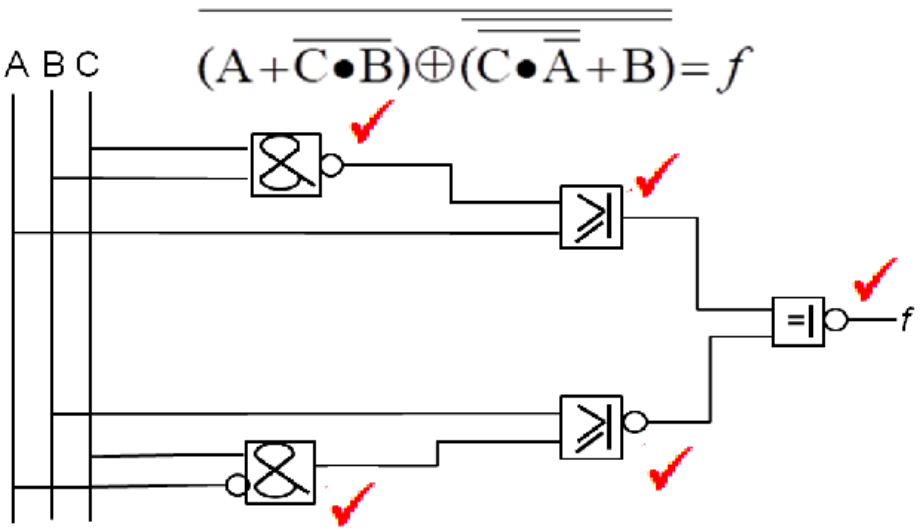
$$\overline{(A \oplus \overline{B \bullet C}) \oplus C \bullet (\overline{A} \bullet \overline{B + \overline{C}})} = f$$



(6)

5.2

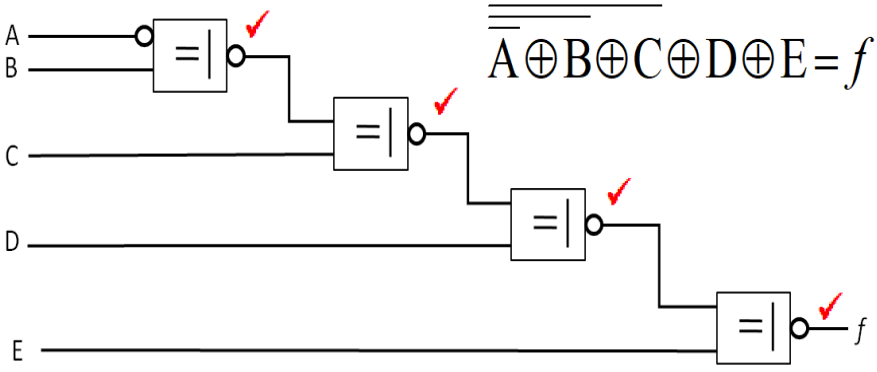
$$\overline{(A + \overline{C \bullet B}) \oplus \overline{(C \bullet \overline{A} + B)}} = f$$



(5)

5.3

$$\overline{\overline{A} \oplus B \oplus C \oplus D \oplus E} = f$$



(4)
[15]

QUESTION 6

6.1

A	B	C	O/P✓✓
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(2)

6.2 $\overline{A} \bullet \overline{B} \bullet \overline{C} + \overline{A} \bullet B \bullet C + A \bullet \overline{B} \bullet C + A \bullet B \bullet \overline{C} =$ O/P✓

$= (\overline{A} \bullet (\overline{B} \bullet \overline{C}) + (B \bullet C)) + (A \bullet (\overline{B} \bullet C) + (B \bullet \overline{C}))$✓

$= \overline{A} \bullet (\overline{B \oplus C}) + A(B \oplus C)$✓

LET X = $B \oplus C$

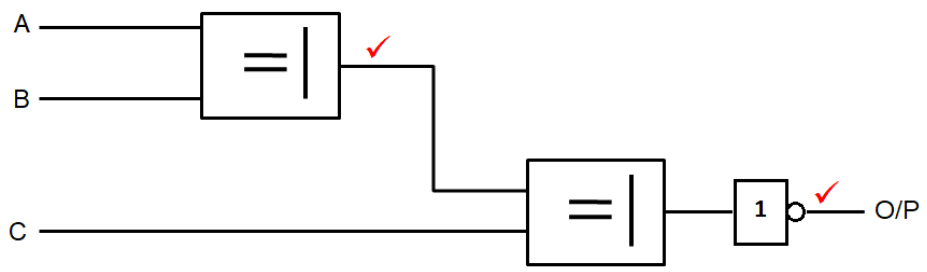
$= \overline{AX} + AX$ ✓

$= \overline{A \oplus X}$ ✓

$= \overline{A \oplus B \oplus C}$ ✓

(6)

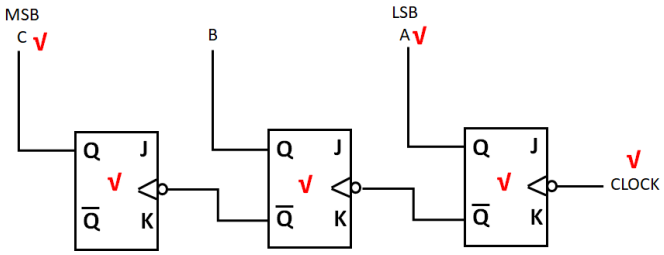
6.3

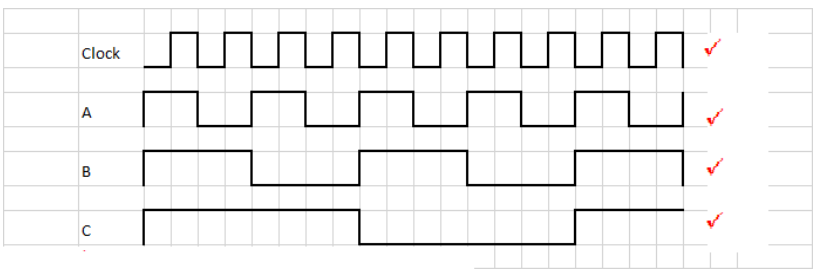


(2)
[10]

QUESTION 7

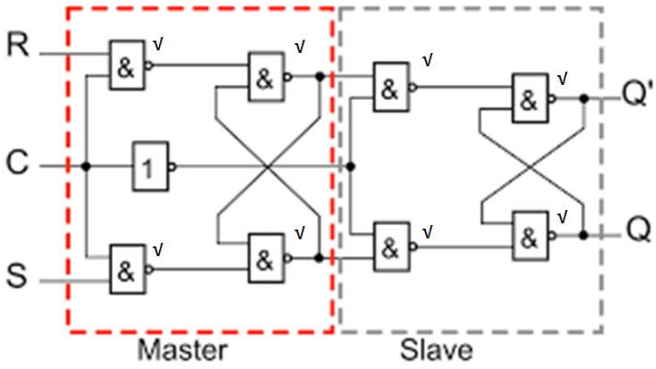
7.1





(10)

7.2



(8)

7.3
- If the set or the pre-set line is activated by changing it to a 0, Q will become a 1 and $\overline{Q}$ will become a 0.
- If the reset or the clear line is activated by changing it to a 0, $\overline{Q}$ will become a 1 and Q will become a 0. (2 × 2) (4)

7.4
- When the clock on the master is a 1, the clock on the slave will be a 0.
- When the clock on the master is a 0, the clock on the slave will be a 1.
- To be able to change Q and $\overline{Q}$, the clock on the slave must change from a 0 to a 1.
- Therefore, the clock on the master will have to change from a 1 to a 0 for the clock on the slave to change from a 0 to a 1. (4 × 2) (8)

[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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